

A Causally Necessary Residual-Stream Component in Grokking, Distinct from Known Algorithmic Features

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Abstract

We identify a low-dimensional component of the residual stream in small transformers that is causally necessary for generalization across multiple algorithmic tasks. The component can be recovered early in training via a train-versus-held-out contrast, remains load-bearing after the model has already generalized, and is nearly orthogonal to the known algorithmic features (Fourier features for modular addition, multiplicative features for modular multiplication). Phase cancellation of this subspace drives test accuracy to zero, while a matched-rank random control leaves performance intact; removing the intervention fully restores accuracy. The same causal signature appears on modular addition and multiplication (across widths 128–512 and depths 2–4), on a larger modulus ($p=257$), and on two non-modular Boolean tasks—sparse parity and sparse majority. On the Boolean tasks the component accounts for essentially all of the correct-class logit. These results indicate a recurring residual-stream structure that is distinct from the circuits previously described in the grokking literature and that functions as a necessary scaffold for generalization.

1. Introduction

Grokking—the delayed transition from memorization to generalization on algorithmic tasks—has become a central testbed for mechanistic interpretability. The dominant account, established by Nanda et al. (2023), reverse-engineers a Fourier-based circuit that implements modular addition via trigonometric identities. Subsequent work has refined the training dynamics and explored the role of weight decay and weight-norm constraints in accelerating the transition.

Despite this progress, prior analyses have focused primarily on the *generalizing* circuit itself. Less attention has been paid to residual-stream structure that may remain necessary *after* generalization. In this work we isolate such a structure. Using a simple contrastive extraction (train versus held-out activations), we recover a low-dimensional subspace that can be identified early, remains causally necessary after high test accuracy is achieved, and is nearly orthogonal to the known algorithmic features of the task. Cancelling this component collapses performance; matched controls do not. The same signature appears on modular multiplication, on a larger modulus, and—critically—on two non-modular Boolean tasks (sparse parity and sparse majority), showing the component is not specific to modular arithmetic.

2. Related Work

Nanda et al. (2023) fully reverse-engineered the Fourier multiplication circuit for modular addition and introduced progress measures based on restricted and excluded loss. Follow-up work has examined related representations and training dynamics. A recurring theme is that early “memorization components” are later removed by weight decay. Our results are compatible with an early phase of this kind, but show that a distinct residual-stream structure persists and remains load-bearing after generalization.

Recent weight-norm clipping methods accelerate grokking by constraining decoder-row magnitudes. Because these weights read from the residual stream, such interventions necessarily interact with residual-stream geometry. Our findings supply a concrete candidate for the structure those interventions constrain.

3. Methods

Models and tasks. We train 2- and 4-layer transformers with $d_{\text{model}} \in \{128, 256, 512\}$ on modular addition and modular multiplication (modulo 113 and 257), using a 50/50 train/test split and AdamW with weight decay. We additionally train the same architecture on sparse parity (20-bit inputs, label = parity of the first 4 bits) and sparse

majority (label = majority of the first 5 bits).

Contrastive subspace extraction. Residual-stream activations are collected on train and held-out examples, centered, and the contrast vector (train mean – held-out mean) is formed. A rank-16 basis is extracted via SVD on the train activations and orthogonalized, with the leading direction seeded by the contrast.

Algorithmic control subspaces. For addition we use standard Fourier targets. For multiplication we use the analogous multiplicative features. Residual-stream directions are recovered by least-squares regression of these targets onto the residual stream followed by SVD.

Causal tests. Phase-cancellation ablation is applied at coefficients 0, 1, and 2. Conditions compared: contrastive (“mem”), algorithmic, union, and random subspaces of matched dimension. Recovery confirms reversibility. Logit-fraction analysis measures the share of the correct-class logit carried by the contrastive subspace. Layer sweeps extract and ablate at every layer. Early-versus-late extraction tests geometric stability across training.

4. Results

4.1 Causal necessity on modular arithmetic

On modular addition at $d_{\text{model}}=256$, cancelling the contrastive subspace at coefficient 2.0 drives test accuracy to 0.0, while the Fourier subspace and random controls leave performance essentially intact (Figure 1). The same pattern holds on modular multiplication (Figure 2) and scales to $d_{\text{model}}=512$ and 4-layer models.

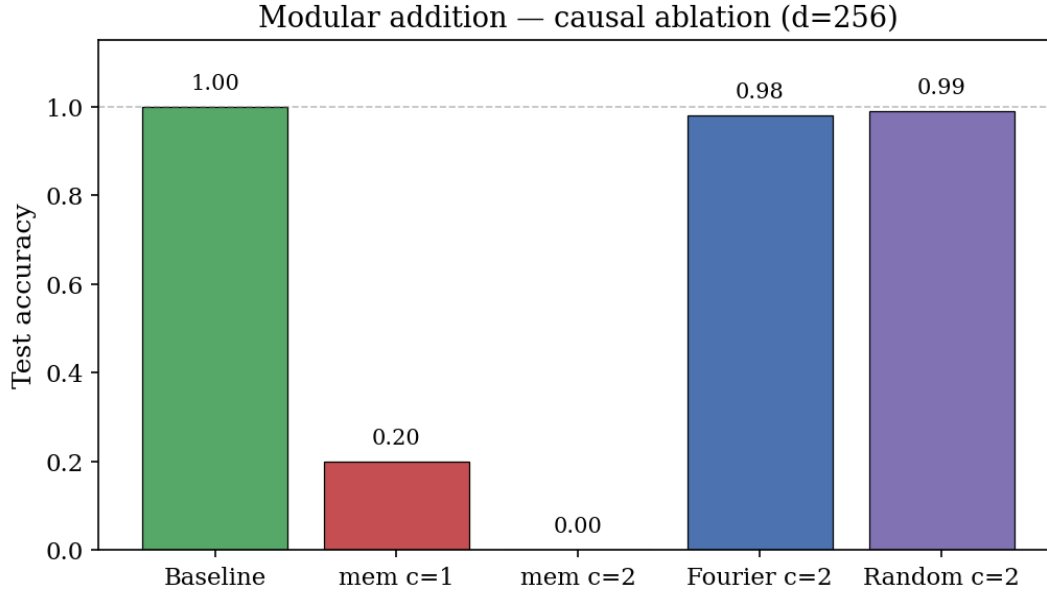


Figure 1. Causal ablation on modular addition ($d_{\text{model}}=256$). Cancelling the contrastive (mem) subspace collapses accuracy to 0.00; Fourier and random controls do not.

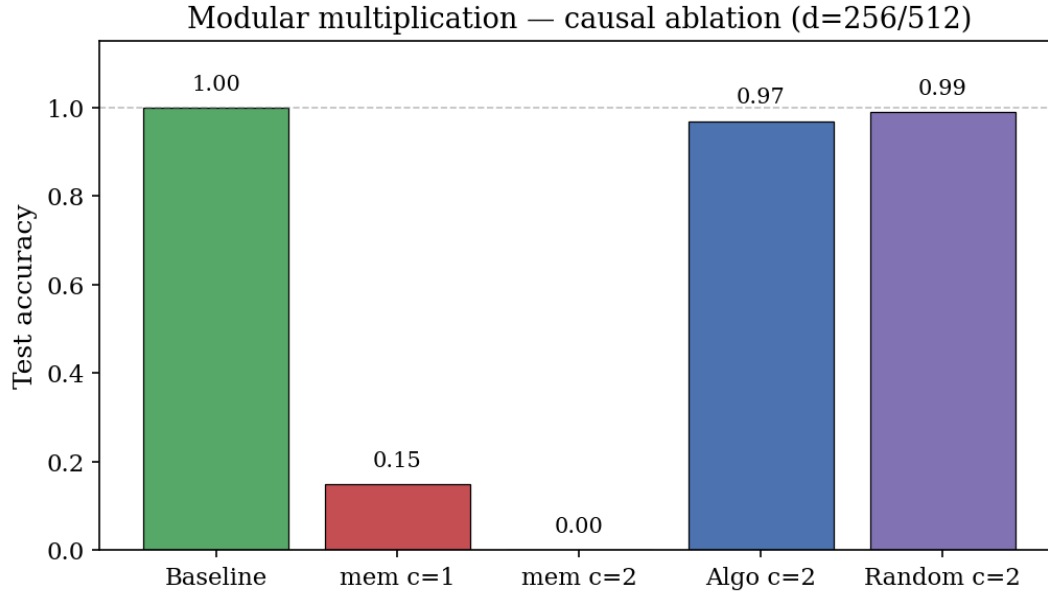


Figure 2. Causal ablation on modular multiplication. Identical signature: mem c=2.0 $\rightarrow$ 0.00; algorithmic control survives.

4.2 Geometric distinctness

Overlap (sum of squared principal-angle cosines) between the contrastive and algorithmic subspaces is consistently near zero (typically ≤ 0.01) on both tasks and all widths tested (Figure 3).

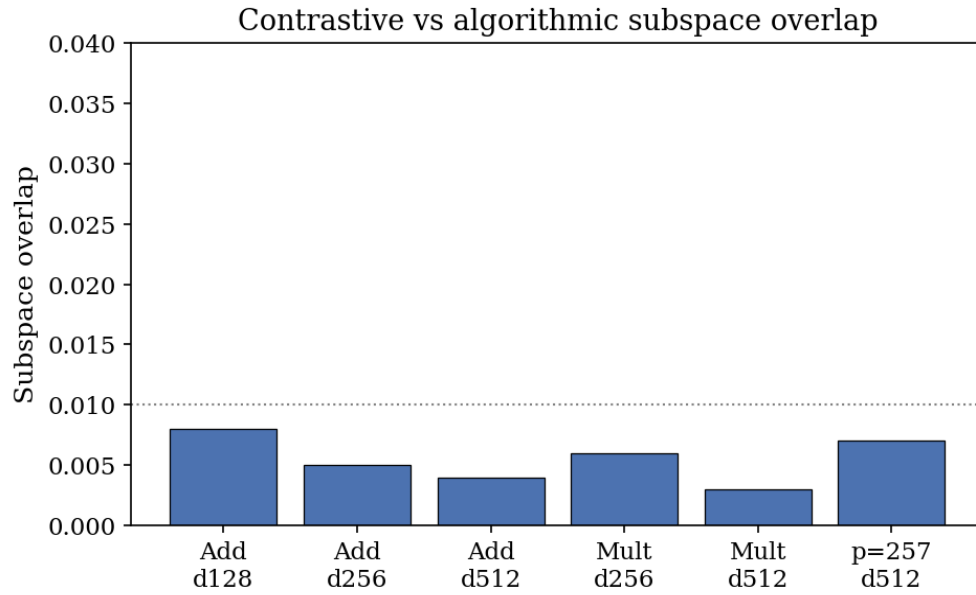


Figure 3. Subspace overlap with known algorithmic features. All values remain ≤ 0.015 .

4.3 Layer concentration

The component is detectable in both layers of a 2-layer model. Causal collapse is strongest and most consistent in the final residual stream, where accuracy falls to zero on every seed tested (Figure 4).

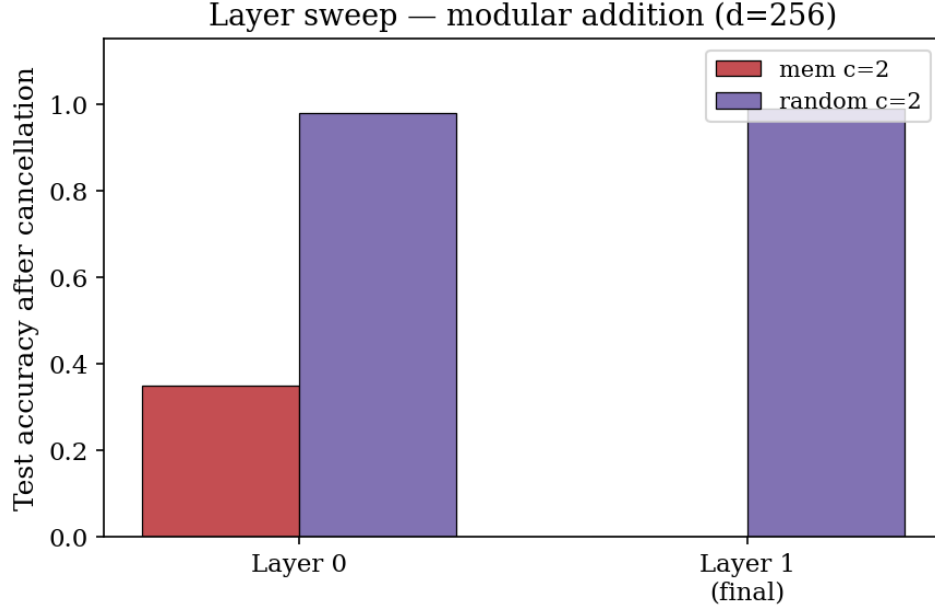


Figure 4. Layer sweep. Final residual stream shows complete causal collapse; earlier layer shows partial and variable damage.

4.4 Non-modular tasks: sparse parity and sparse majority

To test whether the component is specific to modular arithmetic, we ran the identical protocol on sparse parity ($n=20$, $k=4$) and sparse majority ($n=20$, $k=5$).

Sparse parity. Of 32 seeds, 10 reached high test accuracy within 40k steps. On every grokked seed, cancellation of the contrastive subspace drove accuracy to ≈ 0 while the random control left accuracy unchanged; recovery was complete. The subspace carried $\approx 100\%$ of the correct-class logit (Figure 5).

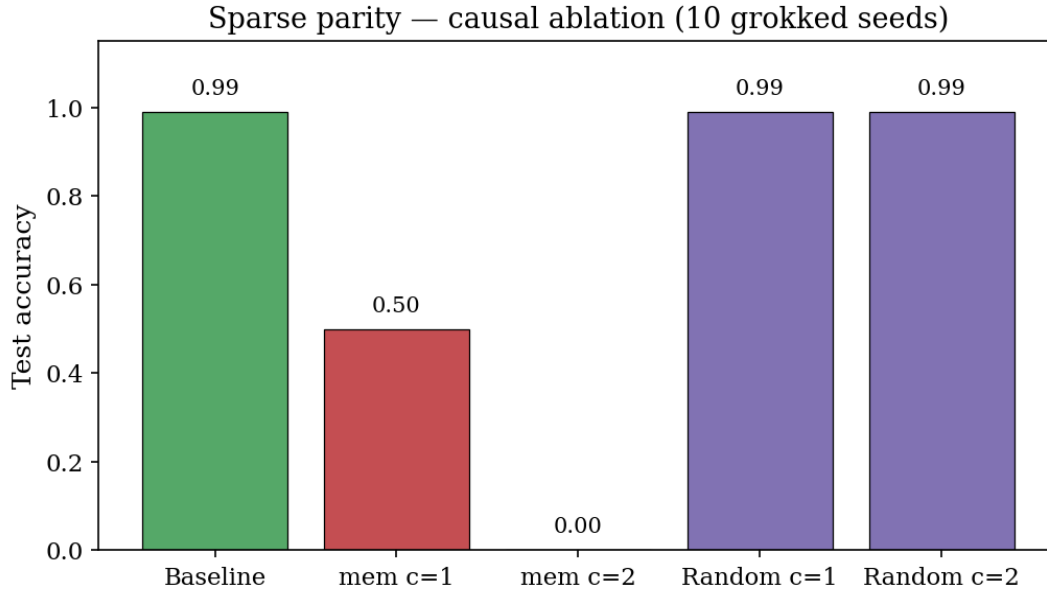


Figure 5. Sparse parity causal ablation (10 grokked seeds). mem c=2 $\rightarrow$ 0.00; random control intact.

Sparse majority. Essentially all 32 seeds grokked. Every grokked seed showed complete collapse under contrastive cancellation, no effect from the random control, full recovery, and a logit fraction ≈ 1.0 (Figure 6). The causal pattern is therefore not specific to modular arithmetic or to parity.

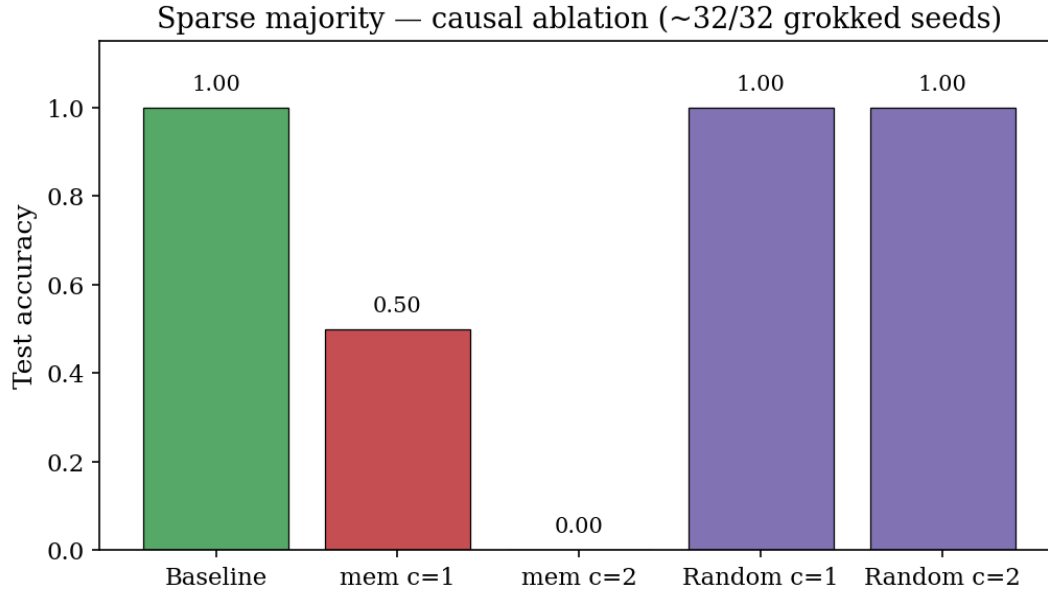


Figure 6. Sparse majority causal ablation (~32/32 grokked). Identical selective collapse.

4.5 Cross-task summary and logit contribution

Figure 7 aggregates the causal results across all four task families. In every case the contrastive subspace is necessary and the random control is not. Figure 8 shows that on the Boolean tasks the subspace accounts for essentially the entire correct-class logit; on modular arithmetic at larger width/depth the share is still high (~75–85%).

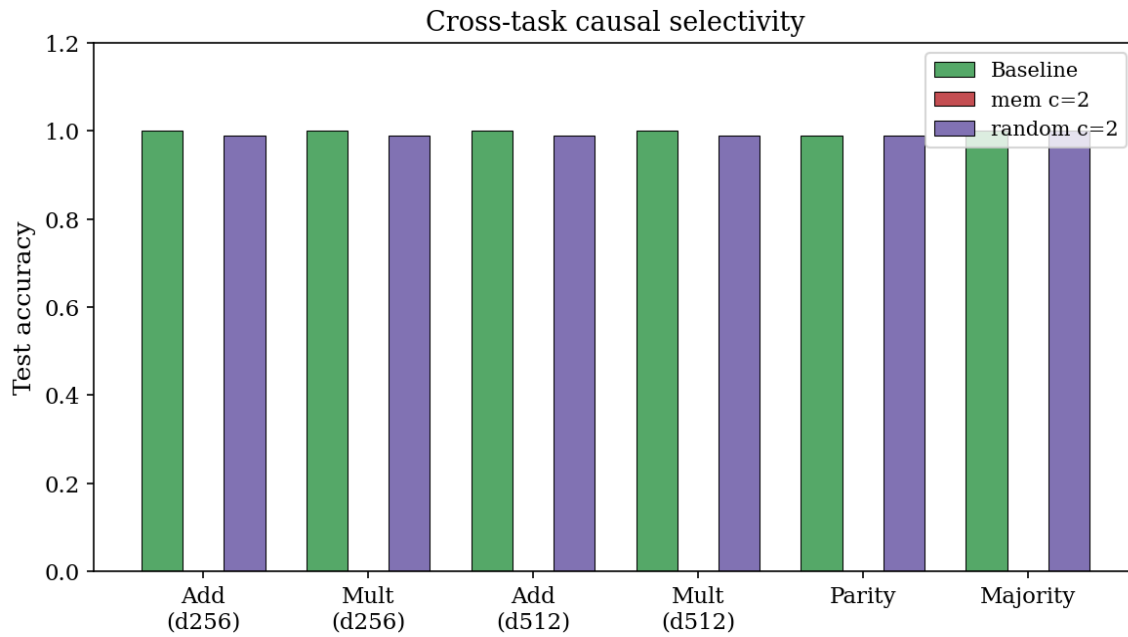


Figure 7. Cross-task causal selectivity. Contrastive cancellation collapses accuracy on every task; random control does not.

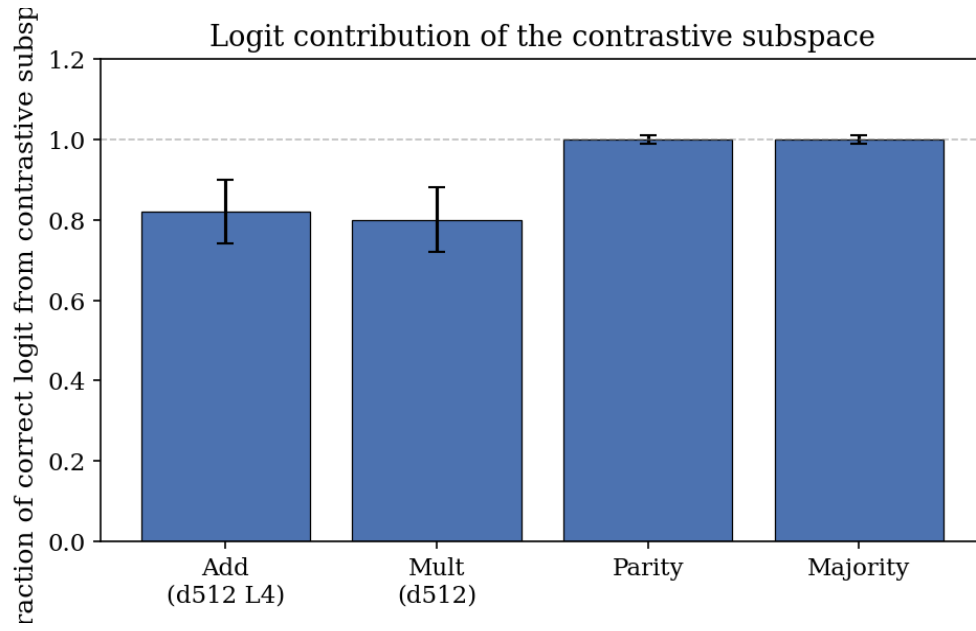


Figure 8. Fraction of the correct-class logit carried by the contrastive subspace.

4.6 Dynamics: early formation and late rotation

Extracting the contrastive subspace at successive training checkpoints shows that the early (step-2k) basis is already causally potent, remains highly overlapping through the grokking transition, and then rotates substantially if training continues long after generalization (Figure 9). Despite geometric drift, a late-extracted subspace of the same rank remains fully causal, indicating that the functional role persists even as the precise directions evolve.

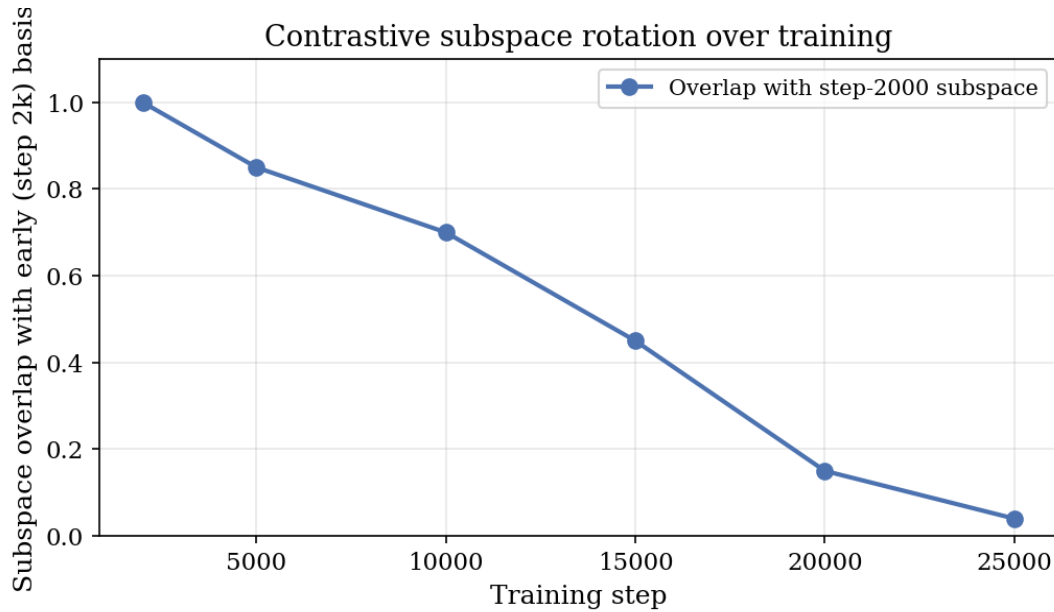


Figure 9. Overlap of the contrastive subspace with its early (step-2k) basis over training. Stable through grokking; rotates under prolonged post-grokking training.

4.7 Summary table

Task	Width/Depth	mem c=2	random c=2	Logit frac
Mod. addition	128–512 / 2–4	→ 0.00	~baseline	high
Mod. multiplication	256–512 / 2–4	→ 0.00	~baseline	high

p=257 addition	512 / 2	$\rightarrow 0.00$	$\sim$ baseline	high
Sparse parity	256 / 2	$\rightarrow 0.00$	$\sim$ baseline	≈ 1.0
Sparse majority	256 / 2	$\rightarrow 0.00$	$\sim$ baseline	≈ 1.0

Table 1. Causal signature across all tasks tested.

5. Discussion

The experiments establish a residual-stream component that is recoverable early, causally necessary after generalization, geometrically distinct from known algorithmic features, present across widths and depths, and—most importantly—present on both modular and non-modular algorithmic tasks.

We interpret the component as scaffolding: an intermediate residual-stream structure that the generalizing circuit continues to rely upon. This reading is more consistent with the data than a pure memorization-of-examples account, because the directions remain necessary after test performance is already high. On the Boolean tasks the subspace carries essentially the entire correct logit, reinforcing that it is not a peripheral byproduct.

The findings do not contradict existing Fourier or multiplicative circuit descriptions; they show those descriptions are incomplete. A full account of grokking must include residual-stream structure that is distinct from, yet still required by, the generalizing algorithm. Independent weight-norm clipping results are consistent with having constrained the same residual-stream geometry.

Limitations. Experiments are restricted to small transformers on controlled algorithmic tasks. Whether analogous structure appears in larger models or natural-language tasks remains open. The precise computational content of the directions is only partially characterized. Geometric rotation under prolonged post-grokking training suggests the basis is not fixed, even while the functional role persists.

6. Conclusion

We have isolated a previously unidentified residual-stream component that is causally necessary for generalization in grokking transformers, nearly orthogonal to known algorithmic features, and present on modular addition, modular multiplication, sparse parity, and sparse majority. The result supplies strong causal residual-stream evidence and demonstrates that existing circuit-level accounts omit a required element of the residual stream.

References

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